

Sacha Evain

Machine Learning Engineer | Applied AI & MLOps

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Profile

Big Data and Artificial Intelligence engineer, ranked 1st out of 204 students in my Master's year, with hands-on experience building data pipelines, predictive models, internal data tooling, and AI-enabled systems in an industrial environment. Seeking a permanent role to deliver reliable data and machine learning solutions with measurable business value.

Experience

Data & Machine Learning Engineer — Apprenticeship, Roquette Health & Pharma – Lestrem, France Sept 2025 – Aug 2026

- Designed predictive models for direct compression manufacturing, including production-defect prediction and tablet-quality attribute estimation.
- Built experimental-data pipelines with Airflow, Snowflake, and Python to structure data for model training and evaluation.
- Deployed and tracked ML experiments with Azure ML and MLflow, covering model versioning, experiment traceability, and training orchestration.
- Automated scientific report generation, reducing preparation time from approximately one day to one minute.
- Built a text-to-SQL agent on Snowflake and a retrieval-augmented generation system over internal technical documentation with source citations.

Quantitative Finance Research Assistant, University of Modena and Reggio Emilia – Modena, Italy May 2025 – Aug 2025

- Studied the greenium and analyzed relationships between climate indicators, market data, risk, and asset valuation.
- Developed and compared machine learning and deep learning models on European financial data.

Cybersecurity Analyst, Lenglet-Imprimeurs – Raillencourt, France July 2022 – Aug 2022

- Analyzed an industrial network infrastructure and drafted a business continuity plan to improve operational resilience.

Technical Skills

Data & ML: Python, SQL, PyTorch, Keras, scikit-learn, LightGBM, XGBoost, deep learning, feature engineering, model evaluation, hyperparameter optimization, statistical modeling, forecasting, explainable AI, LLM fine-tuning

Data platforms: Snowflake, Hadoop, Spark, Polars, ETL/ELT, Airflow, Prefect, data quality, data preprocessing, scientific data analysis, data pipeline design

MLOps & infrastructure: Azure ML, MLflow, Weights & Biases, DVC, Docker, Git, Kubernetes, k3s, GitOps, model deployment, experiment tracking, model registry, reproducible ML, ML pipeline automation, Linux

AI systems: LLMs, RAG, text-to-SQL agents, source-grounded generation

Selected Projects

- End-to-end design, training and fine-tuning of LLMs.
- HomeLab infrastructure: self-hosted k3s Kubernetes cluster on Proxmox, GitOps deployments, and server administration.
- Data competitions: Kaggle and CrunchDAO.
- Cybersecurity: Top 500 on Root-Me among approximately 300,000 participants — 7,500 points.

Education

Junia ISEN Lille, Engineering degree in Big Data & Artificial Intelligence – Lille, France 2021 – 2026

- Ranked 1st out of 204 students in the Master's year; coursework included machine learning, deep learning, data analysis, algorithms, metaheuristics, Hadoop, and Spark.

Languages

English: C1 — TOEIC 885

French: Native

Italian: B1